

SINGAPORE PAYA LEBAR, Singapore

WMO: 486940

Lat: 1.360N

Lon: 103.910E

Elev: 65

StdP: 14.66

Time Zone: 8.00 (E08)

Period: 99-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	74.8	75.1	67.9	102.7	84.0	69.4	108.5	83.3	20.4	85.2	18.8	85.3	4.8	10	0.380

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5	10.7	93.5	79.4	92.9	79.4	91.6	79.1	82.6	88.0	81.8	87.3	81.3	86.9	7.6	60

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
80.9	160.9	86.6	80.5	158.9	86.2	79.2	152.0	84.8	46.3	89.7	45.6	88.2	45.0	87.5	90.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.5	14.5	12.8	72.6	95.5	1.2	1.1	71.7	96.3	71.0	96.9	70.3	97.6	69.4	98.4
			69.8	84.1	1.4	1.9	68.8	85.5	68.0	86.6	67.3	87.6	66.3	89.0
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	83.9	82.3	83.2	84.2	85.0	85.3	84.7	83.9	83.8	84.0	84.2	83.5	82.8
	DBStd	2.32	2.39	1.90	2.16	2.12	2.09	2.34	2.30	2.25	2.11	2.12	1.86	2.20
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD50	12377	1001	929	1059	1049	1095	1041	1052	1049	1019	1061	1006	1016
	CDD65	6902	536	509	594	599	630	591	587	584	569	596	556	551
	CDH74	78235	5317	5281	6549	6968	7822	7313	6965	6911	6699	6910	5917	5583
	CDH80	30083	1634	1809	2506	2891	3490	3147	2767	2693	2636	2754	2019	1737
Wind	WSAvg	5.5	7.6	8.2	6.0	3.7	4.2	5.0	5.9	6.0	5.4	4.3	3.8	5.9
Precipitation	PrecAvg	84.8	7.1	6.9	6.3	5.9	6.1	5.7	5.7	5.6	6.3	6.9	9.2	12.8
	PrecMax	116.3	19.0	18.5	14.2	13.6	10.4	10.0	10.0	10.2	13.5	17.0	18.3	26.4
	PrecMin	61.5	0.9	0.2	0.7	0.7	3.0	2.3	1.1	2.0	1.9	1.5	2.6	4.9
	PrecStd	16.9	4.7	5.0	3.6	3.3	1.7	2.1	2.1	2.1	2.7	4.2	3.9	6.1
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	91.8	93.1	93.6	95.1	94.9	93.5	91.7	91.8	92.9	93.6	93.2	93.0
		MCWB	79.0	79.5	79.4	80.0	79.8	78.4	78.8	78.2	78.4	78.1	79.5	79.1
	2%	DB	90.0	91.4	92.8	93.3	93.0	91.7	91.0	91.0	91.3	91.7	91.4	91.1
		MCWB	78.5	79.0	79.4	80.3	79.8	78.9	78.7	78.4	78.3	78.5	79.2	78.9
	5%	DB	89.2	89.8	91.3	91.6	91.5	91.0	89.6	89.6	89.8	91.0	89.8	89.4
		MCWB	78.4	78.2	78.9	79.9	79.8	79.0	78.6	78.5	78.2	78.5	78.8	78.8
	10%	DB	87.6	88.1	89.6	89.9	89.8	89.5	88.1	88.0	88.2	89.4	88.1	87.6
		MCWB	78.2	78.1	78.6	79.6	79.8	79.0	78.9	78.6	78.6	78.4	79.1	78.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	82.5	83.1	82.1	83.3	83.3	82.3	82.4	82.3	81.6	82.2	82.5	82.1
		MCDB	87.8	89.0	88.9	89.8	89.0	86.9	86.4	86.7	86.4	87.4	87.5	87.0
	2%	WB	80.3	80.7	81.0	82.0	82.0	81.4	81.0	80.9	80.5	80.9	80.9	80.7
		MCDB	86.2	87.0	87.8	88.4	87.5	86.4	85.8	85.9	86.2	86.6	86.8	86.6
	5%	WB	79.6	79.7	80.2	81.2	81.4	80.7	80.3	80.1	80.0	80.1	80.0	79.9
		MCDB	85.3	86.0	86.8	87.5	87.0	86.1	85.4	85.3	85.7	85.9	85.9	85.6
	10%	WB	78.9	79.0	79.7	80.5	80.8	80.1	79.9	79.7	79.6	79.7	79.6	79.3
		MCDB	84.5	85.2	86.1	86.7	86.5	85.7	85.2	85.0	85.3	85.5	85.3	84.7
Mean Daily Temperature Range	5% DB	MDBR	10.3	11.7	12.2	12.2	10.7	9.8	9.4	9.3	10.3	11.2	11.3	10.5
		MCDBR	12.4	13.1	13.6	13.5	11.9	10.9	10.1	10.3	11.2	12.5	12.8	12.8
		MCWBR	4.8	4.6	4.8	4.9	4.7	4.5	4.4	4.4	4.7	4.7	5.0	4.9
	5% WB	MCDBR	11.3	12.0	12.5	12.3	10.8	9.9	9.4	9.6	10.4	11.1	11.6	11.3
		MCWBR	5.1	5.3	5.3	5.5	5.1	4.7	4.8	4.8	5.0	4.9	5.1	5.1
Clear-Sky Solar Irradiance	taub	0.470	0.482	0.492	0.489	0.475	0.486	0.478	0.521	0.552	0.579	0.484	0.470	
	taud	2.297	2.244	2.226	2.234	2.282	2.253	2.283	2.141	2.050	1.955	2.254	2.304	
	Ebn at Noon	273	272	267	262	259	252	255	250	248	243	267	271	
	Edh at Noon	43	46	47	46	42	43	42	49	56	61	45	43	
All-Sky Solar Radiation	RadAvg	1441	1667	1661	1609	1487	1410	1401	1424	1421	1420	1343	1311	
	RadStd	121	149	146	102	71	101	107	81	97	105	84	98	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+1.03	+0.49	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (6 neighbors)	+0.67	+0.16	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page